

Saanvi Skanda Subramanian

Christopher Hitchcock

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Transformative Experiences Change Our Brains

The Stanford Encyclopedia of Philosophy defines transformative experiences as experiences that radically change the experiencer in both an epistemic and personal way (*Transformative Experience (Stanford Encyclopedia of Philosophy)*, n.d.). An experience is personally transformative if it changes what it is like to be you: after undergoing a personally transformative experience, your conscious experiences are unlike any previous experiences. Experiences are epistemically transformative in that they impart knowledge impossible to acquire any other way. Together, these two classifications bring about fundamental changes in what a person values. In the précis of her book *Transformative Experiences*, Paul offers several examples: becoming a parent, undergoing a religious conversion, fighting in a war, etc. In this paper, I will argue that transformative experiences do turn you into a different person because not only has the sense of self fundamentally changed, but there are measurable neurobiological differences in the body as well. The neurobiological evidence provides a unique angle to Paul's philosophic argument and extends it. Paul argues that transformative experiences change the self by altering values and what it is like to be you. The biological evidence suggests that these changes are physically instantiated in the brain and body, which provides a measurable basis for claiming that the person who emerges from a transformative experience is genuinely a different person.

Paul defines a transformative experience as a “kind of experience that is both radically new to the agent and changes her in a deep and fundamental way” (Paul, 2015). In framing her argument, Paul draws a distinction between a “person” and “self” which is important in grounding the argument (*Transformative Experience* | *Oxford Academic*, n.d.). While one generally remains the same person so long as they retain the same brain and body, Paul claims that the self is more fragile. This is because one can undergo transformations so radical that they give rise to an entirely new self, without this constituting death or the end of the person. Importantly, Paul argues that transformative experiences are impossible to rationally anticipate in advance. This implies that rational decision-making requires understanding what outcomes are desirable, although transformative experiences can change the neural basis for the decision-making. The decision to undergo a transformative experience is therefore a *leap of faith*. This is where understanding neurobiological evidence becomes significant because if the brain is structurally reorganized from these experiences, then the production of a new self becomes more grounded through biological, quantifiable evidence. Moreover, the neurobiological dimension addresses a gap in Paul’s argument—she focuses on what it is like to be you in a phenomenological sense, but this neurobiological evidence suggests that the *processor* that generates subjective experiences has been physically altered. What it is like to be you changes precisely because the organ undergoing the experiencing has been reorganized, and this is the basis of my argument.

Current neurobiological research has examined near-death experiences, encounter-type experiences and out-of-body experiences at both the psychological and physiological levels. Taking combat as an example, we can look directly at the neurological effects of severe trauma. Research shows that traumatic stress affects several key brain regions like the amygdala,

hippocampus, and prefrontal cortex, and can produce lasting structural and functional changes in each. Traumatic stress is also associated with elevated cortisol and norepinephrine responses to subsequent stressors. These hormonal changes are significant in personal transformation because they rewire the body's threat response, and ordinary situations like sleep and everyday sounds can be experienced through fundamentally different processing. This would suggest that veterans are shaped by an altered nervous system that processes the world differently at every moment. Findings from animal studies have been extended to patients with post-traumatic stress disorder (PTSD), which reveals smaller hippocampal and anterior cingulate volumes, heightened amygdala reactivity, and decreased medial prefrontal cingulate functioning. Notably, effective PTSD treatments have been shown to promote neurogenesis through improving memory consolidation and increasing hippocampal volume (*Traumatic Stress: Effects on the Brain - PMC*, n.d.). This suggests that the brain is actively reorganized from the experience as opposed to mere damage. These changes are not transient or temporary, and they represent alterations to neural architecture, which supports that a person who has fought in a war and experienced PTSD as veterans do is biologically not the same person who enlisted.

Becoming a parent offers another well-documented example of neurobiological transformation. A study of more than 100 new parents found that pregnancy and childbirth produce changes in brain circuits associated with empathy and parent-infant bonding. A 2016 study found that gray matter decreased in first-time mothers following birth, with sustained reductions that persist for at least six years postpartum. Researchers interpret these neural reductions as a form of *synaptic pruning*, which is where neural circuits streamline towards social and caregiving functions (*Pregnancy and Childbirth Reshape the Brain in Profound, Sometimes Lasting Ways* | *Scientific American*, n.d.). This further demonstrates how the brain

actively reorganizes its architecture to this transformative experience. The perceptual and emotional areas of the brain are processed through these newly formed neural circuits that have been structurally tuned to parenthood. This shows that what it is like to you as a parent is generated by a brain that has physically reshaped itself around the caregiving role. Other studies have observed short-term gray matter increases in the weeks that immediately follow birth. Moreover, fatherhood produces its own neurological shifts. Research conducted jointly by the University of Southern California and the Instituto de Investigacion Sanitaria Gregorio Marañón tracked 40 fathers using MRI scans taken during their partner's pregnancy and again when the child was six months old, alongside a control group of 17 childless men (*Fatherhood Changes Men's Brains, According to before-and-after MRI Scans*, n.d.). Significant changes appeared in fathers' brains in regions governing visual processing, attention, and empathy towards the infant, and these changes were absent in the control group. Both mothers and fathers undergo structural brain changes after having a child, and this suggests that there are biologically embedded responses to life's most significant experiences.

Another example of a transformative experience causing biological change is with dietary transformation. When a vegetarian eats meat for the first time, the digestive system must work harder to break down denser proteins and higher fat contents. This sudden shift in diet can cause mild, temporary side effects like bloating or indigestion, although some individuals experience no physical reaction. While this example does not reflect psychological depth of other aforementioned significant life transitions, this demonstrates that the body registers the change. Although this may not constitute a truly transformative experience in Paul's philosophical sense, this guides towards the principle that changes in the way we live and psychological changes are closely linked to physical changes.

Religious conversion is another major example through which transformative experiences can be studied neurobiologically. Religious experiences in epilepsy patients are an area of interest of much research, and there has been published evidence that suggests hyperreligiosity may be associated with temporal lobe seizures and post-ictal psychotic episodes. One documented case involves a 75-year-old man with epilepsy who underwent a religious conversion and then developed progressively severe religious delusions and symptoms of Capgras syndrome over the following fifteen years (*Religious Conversion in an Older Male with Longstanding Epilepsy* - *ScienceDirect*, n.d.). Brain imaging showed bilateral posterior cortical atrophy, chronic right parieto-occipital encephalomalacia, and right mesial temporal sclerosis, while neuropsychological testing showed a progression from right temporal lobe abnormalities to broader frontal and bilateral dysfunction. The case highlights how a history of seizures, superimposed on sensory deprivation and a progressive impairment of right posterior and bilateral anterior brain function, may have contributed to religious conversion, which was followed by dementia and delusions involving religious content.

One objection to the claim that transformative experiences make you a fundamentally different person comes from John Locke's memory-based theory of personal identity. He argued that personal identity persists through time as long as there is continuity in consciousness, or whether you can remember being your earlier self. With this philosophy, a veteran who returns from war is still the same person who enlisted because they retain memories of their life before combat. However, Locke's framework is limited because the changes in the brain associated with PTSD, parenthood, and religious experience are structurally significant and more than just accumulations of new memories. Locke's argument is further limited by the concept that memory is simply a reconstruction of past experiences and as easily falsifiable. Elizabeth Loftus,

a distinguished professor at the University of California, Irvine, argues that there is potential for memories to be altered or for completely false memories to be planted and our recollections of events and experiences may be subject to manipulation (*How Memory Can Be Manipulated*, with Elizabeth Loftus, PhD, n.d.). For example, we have explored shrinkage in the hippocampus and hyperactivity in the amygdala. When Locke grounds personal identity in the continuity of memory, he assumes that memory is stable and reliable. However, if the organ that is responsible for memory has itself been reorganized, then the memories that persist are being processed by a structurally different brain than the one that originally formed them.

These examples of combatting trauma, parenthood, dietary change, and religious conversion demonstrates how transformative experiences coincide with measurable, lasting changes in the body and brain. The neurobiological evidence reinforces that these experiences do more than just change what a person believes or values (as assumed from the definition of transformative experiences), and alter the physical body in which the changed beliefs and values are held. This is important for understanding personal identity as if the person who has a transformational experience has a structurally different brain or reorganized bodily responses, then it fundamentally changes who the person is after this experience. Therefore, transformative experiences reshape the self that undergoes them and makes the person going through them as a different person.

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